

BrakeBox

User Manual



How to Read This Manual

BrakeBox has three rotary encoders, each combining a turn knob with a push-button. Throughout this manual they are referred to as **ENC 1** (left), **ENC 2** (middle), and **ENC 3** (right), matching their physical left-to-right position on the unit and the columns shown on screen.

Each encoder supports three actions:

	ENC 1 (left)	ENC 2 (middle)	ENC 3 (right)
T	Turn	Turn	Turn
C	Click	Click	Click
L	Long-press	Long-press	Long-press

Wherever a screen's controls are described, this manual uses a **table with one column per encoder and one row per action (T / C / L)**, the same shape as the control-hint bar shown at the bottom of the BrakeBox display. Where a control needs more explanation than a couple of words, the table cell includes that extra detail directly.

Quick inline references look like **1T** (turn encoder 1) or **3C** (click encoder 3), and can be chained, for example "**hold 3L**" or "**press 2C then 1T**".

1. Introduction

BrakeBox is a DIY, standalone brake pedal controller for sim racing. It reads force from a load cell through an HX711 amplifier, reshapes that input through a configurable response curve, and outputs it to your PC as a standard USB joystick axis, replacing a stock potentiometer pedal's linear, unshaped output.

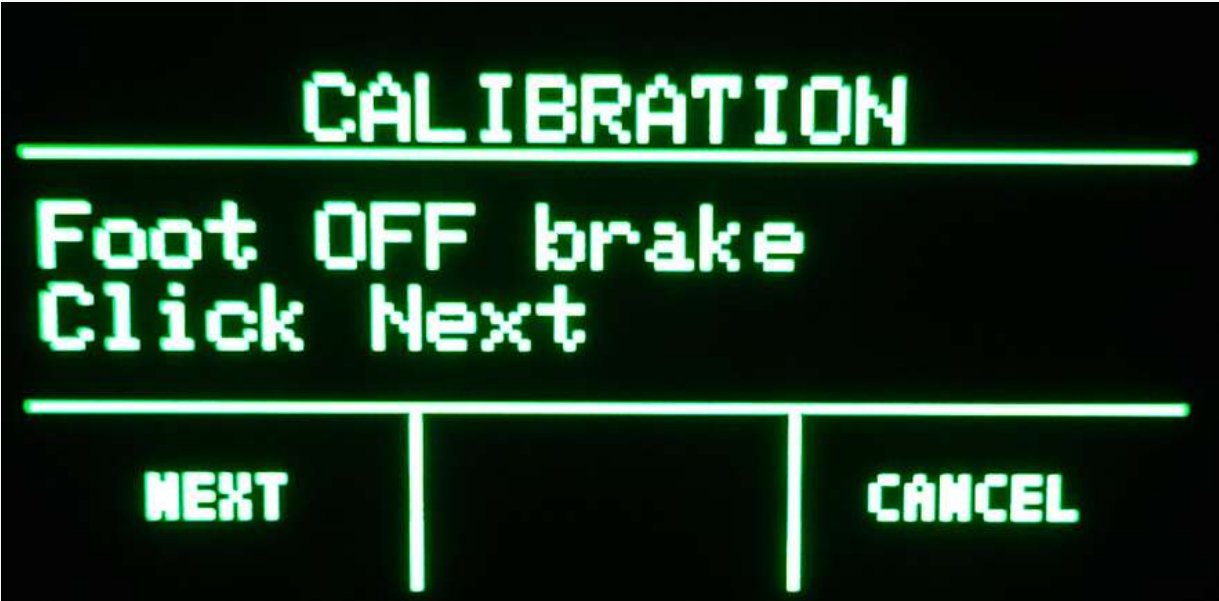
This manual walks through calibration, the main screen, curve shaping, deadzones, and the profile/bank system for saving your setups.

2. Calibration

Before first use, and any time the pedal feels off, run calibration from the main screen: **hold 1L**. You will be prompted to complete calibration after each boot up.

Step 1 — Zero point

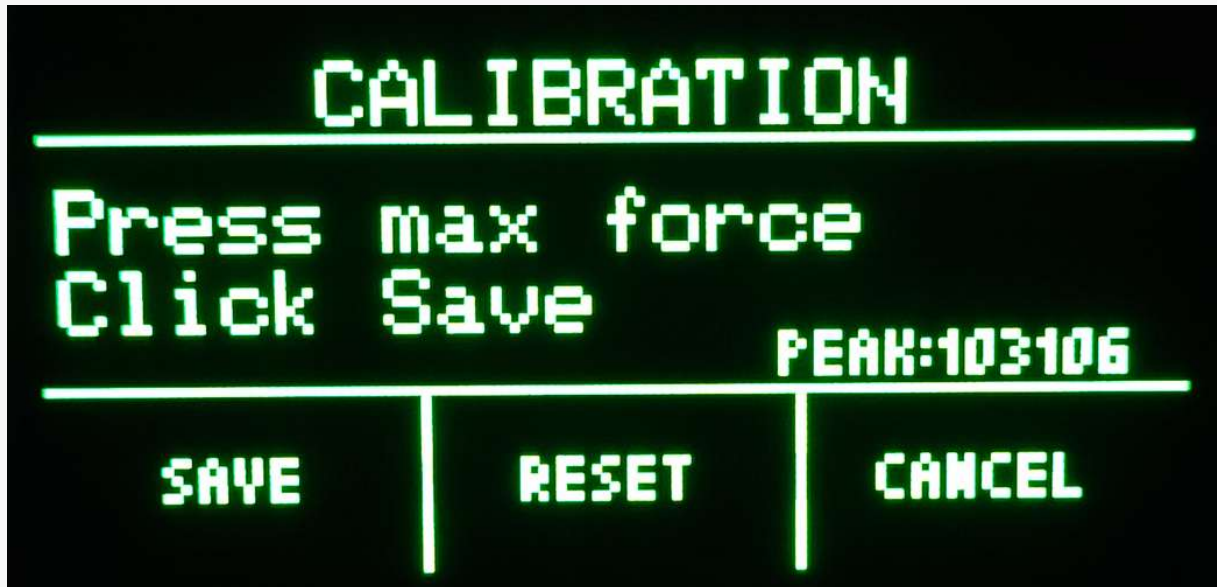
With your foot fully off the pedal, **click 1C** (NEXT) to record the zero-force baseline.



	ENC 1 (left)	ENC 2 (middle)	ENC 3 (right)
T	—	—	—
C	Set baseline value, go to Step 2	—	Abort calibration
L	—	—	—

Step 2 — Max force

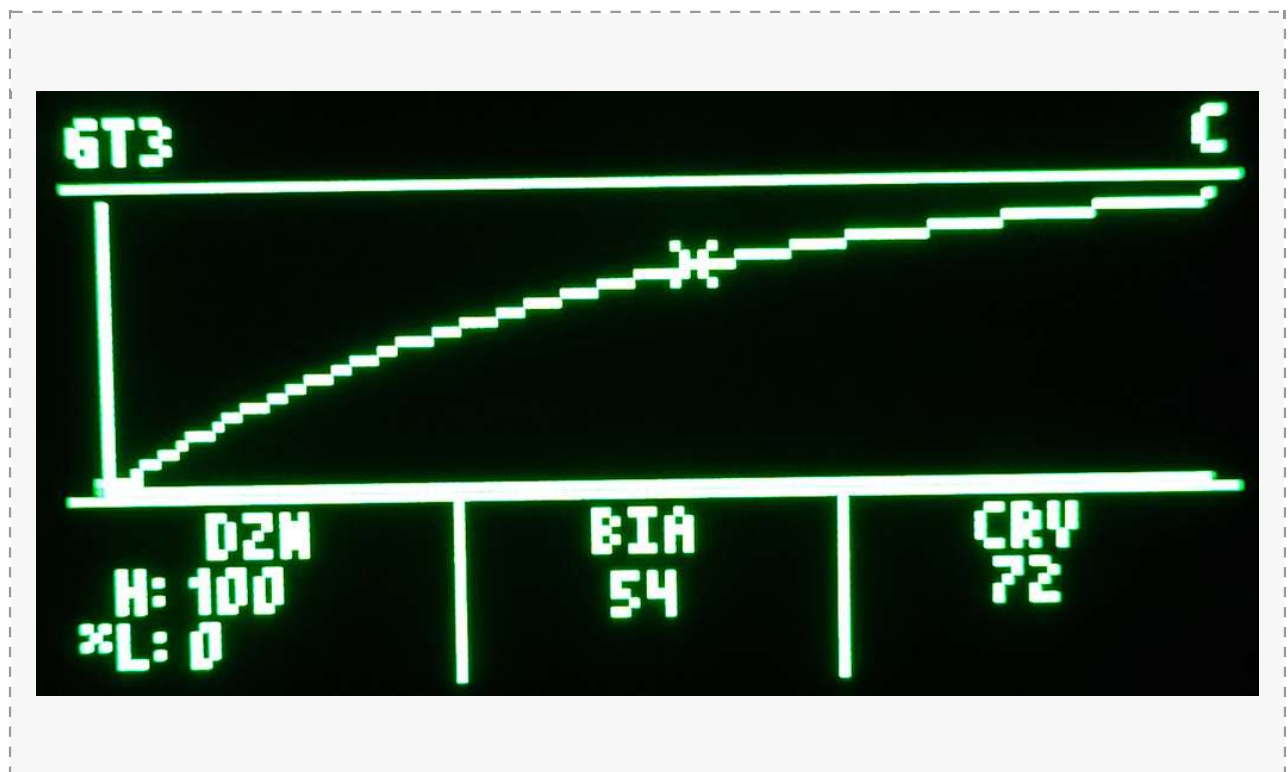
Press the pedal as hard as you comfortably would in real use. BrakeBox will use the highest input recorded for this step; it is not necessary to hold the pedal down while clicking the encoder. The live PEAK reading updates as you press, so you can see the highest force registered so far.



	ENC 1 (left)	ENC 2 (middle)	ENC 3 (right)
T	—	—	—
C	Complete calibration	Set the recorded PEAK back to 0, to redo this step	Abort calibration
L	—	—	—

The calibration prompt shown at power-on can be skipped with 3C (CANCEL). BrakeBox will still function, but the curve will not be tailored to your actual pedal force range until calibration is completed.

3. The Main Screen



The main screen shows your current curve as a graph, with a moving dot marking your live pedal input. The three columns along the bottom, DZN, BIA, and CRV, are controlled by the matching encoder:

	ENC 1 (left)	ENC 2 (middle)	ENC 3 (right)
T	Adjust DZN L/H	Adjust BIA	Adjust CRV
C	Toggle H/L bound	Toggle BIA1/2 (Asym mode) or open Profile Select	Cycle curve mode
L	Enter Calibration	Open Profile Select	Hold-to-save (see Section 6)

For example: to jump straight to calibration, **hold 1L**. To switch curve modes, **click 3C** repeatedly.

The top row shows the current profile name and curve-mode letter, plus flashing warnings for CAL (uncalibrated), SAVE (unsaved changes), or INPUT (no pedal signal) when applicable.

4. Curve Modes

Click **3C** to cycle between three curve modes, shown by the letter in the top-right corner:

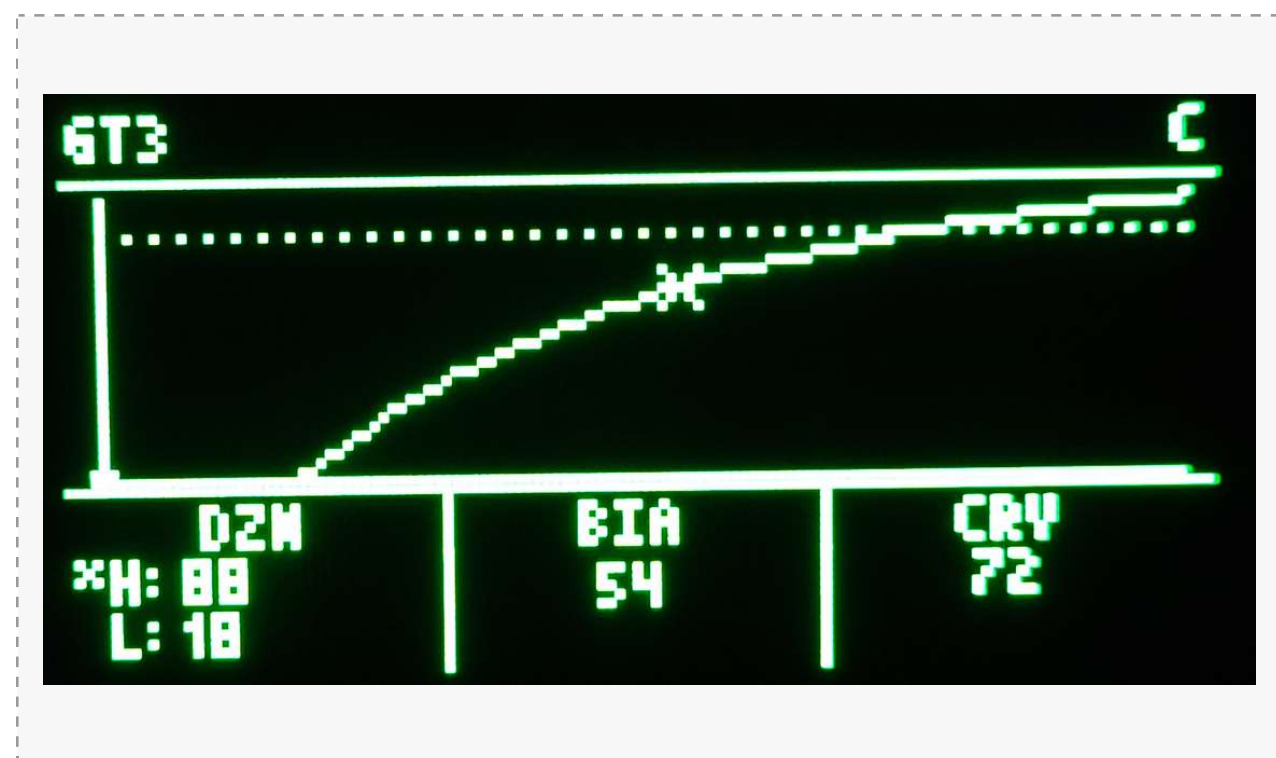
C — Curve. One continuous curve with a single inflection point (BIA).

S — Sym-S. A symmetric S-curve.

A — Asym-S. Two independent S-curves joined at a floating midpoint, giving separate BIA/CRV control over the first and second half of your pedal travel (click **2C** to switch which half you are editing).



5. Deadzones — DZN L and DZN H



L (low) reshapes the curve: input below this threshold reads as zero, and the rest of the curve stretches to fill the remaining travel. Use this to remove a mushy or inconsistent start to your pedal's travel.

H (high) is a soft output cap, not a curve reshape: the full curve is still drawn as if H were 100, but the actual joystick output is clamped at H percent of maximum force. On the graph this shows as a horizontal dotted line; your output can never rise above it. An asterisk (*) appears next to whichever DZN value, L or H, is currently selected for editing.

6. Saving Changes

Changing any parameter causes a flashing SAVE indicator to appear top-right. To save, **hold 3L** on the main screen.

Holding shows a 3-segment progress bar that fills over about 1.5 seconds. Releasing before it fills will abort the save.



PROFILE SAVED!

7. Profiles & Banks

BrakeBox stores up to 500 profiles, organized into 25 banks of 20 profiles each. Bank 0 ("PRESETS") ships with 20 curated factory profiles covering common disciplines (GT3, F1, Rally, Drift, and so on). Banks 1 through 24 are empty and yours to fill and name. Bank 0 and all of its presets can also be renamed, the same as any user bank or profile. To enter Profile/Bank selection press **2C** (when not in A-Symmetric (A) Curve mode) or **hold 2L**.

Profile Select



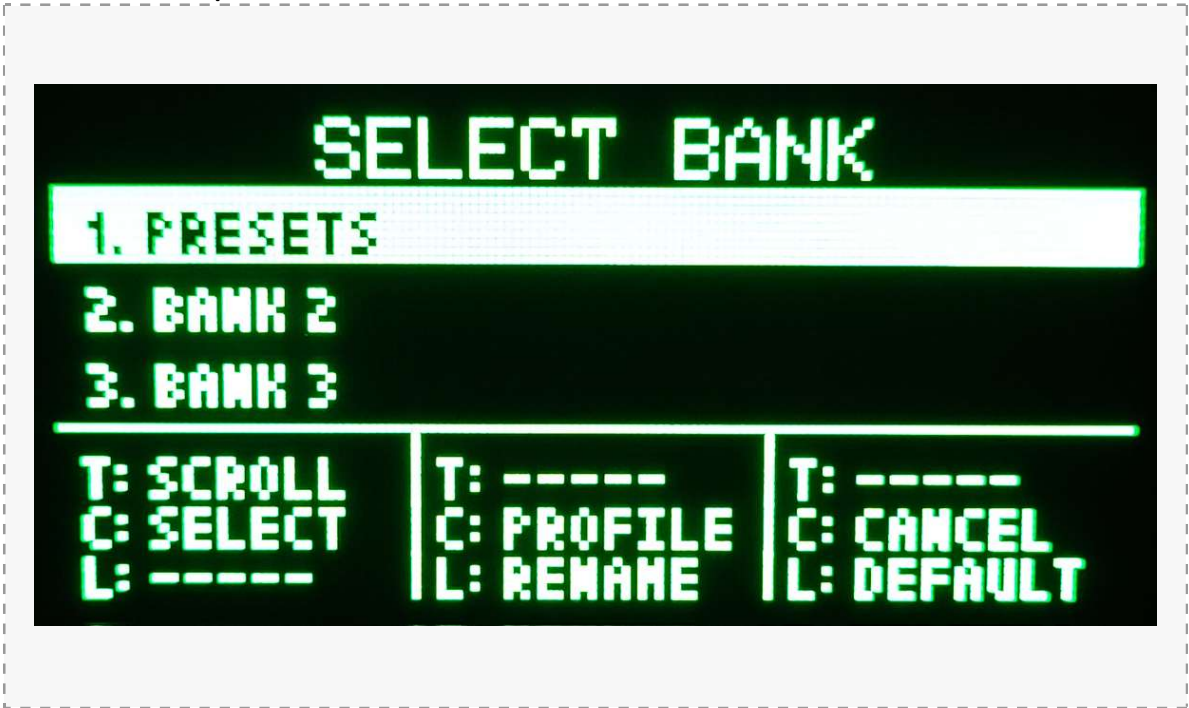
	ENC 1 (left)	ENC 2 (middle)	ENC 3 (right)
T	Scroll	—	—
C	Select	Go to Bank Select	Cancel, return to main screen
L	View description (Bank 0 only)	Rename profile	Reset profile (hold-to-confirm, see Section 6)

Hold 1L to see the description of the in-built profiles (not available on banks 1–24).



Bank Select

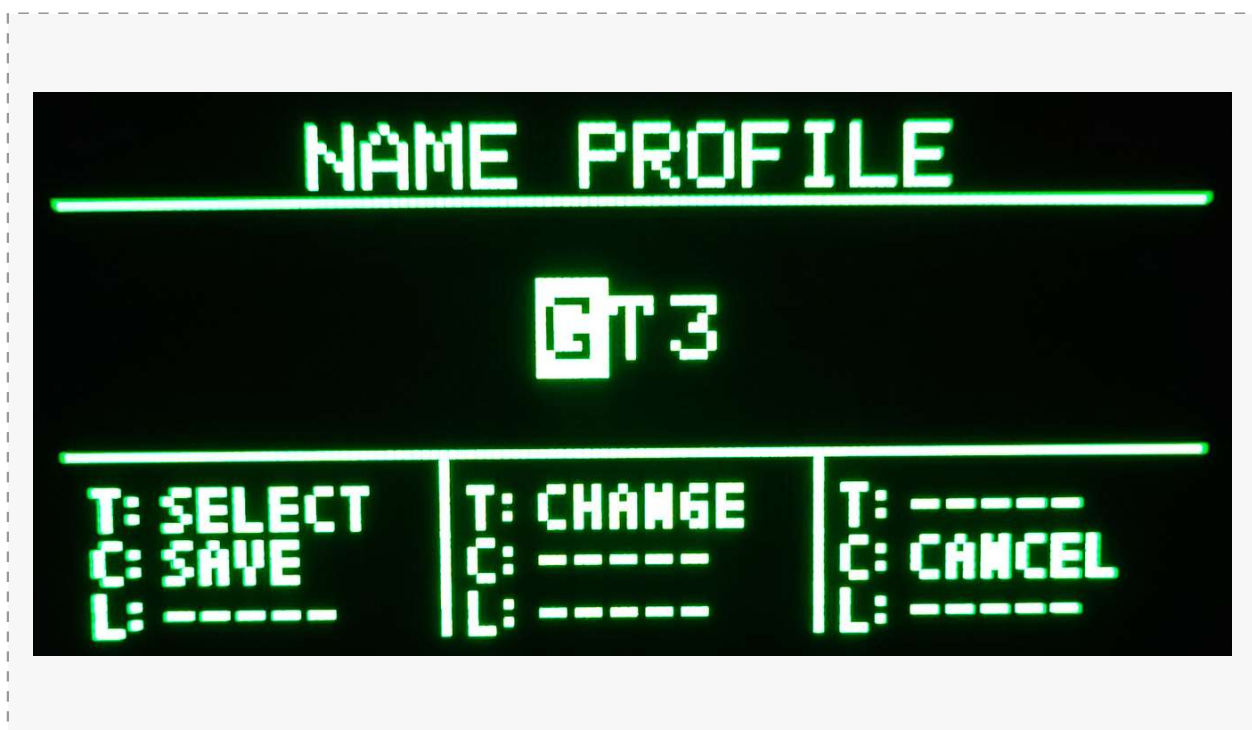
When in Profile Select, **press 2C** to enter Bank Select.



	ENC 1 (left)	ENC 2 (middle)	ENC 3 (right)
T	Scroll	—	—
C	Select bank	Back to Profile Select	Cancel, return to main screen
L	—	Rename bank	Reset bank (hold-to-confirm, see Section 6)

8. Renaming Profiles & Banks

While in either Profile or Bank Select mode, **hold 2L** to rename the profile or bank. Each profile and bank name supports up to 10 characters.



	ENC 1 (left)	ENC 2 (middle)	ENC 3 (right)
T	Move cursor	Cycle alphanumeric	—
C	Save profile/bank name	—	Return to Profile / Bank Select
L	—	—	—

Press 1C to save or **2C** to return to the Profile/Bank Select screen.

9. Factory Reset

A factory reset restores Bank 0 ("PRESETS") back to its original 20 factory profiles. It does not affect your own banks (1–24).

To trigger it, hold the **ENC 3** button down while powering on the unit. A confirmation prompt appears:

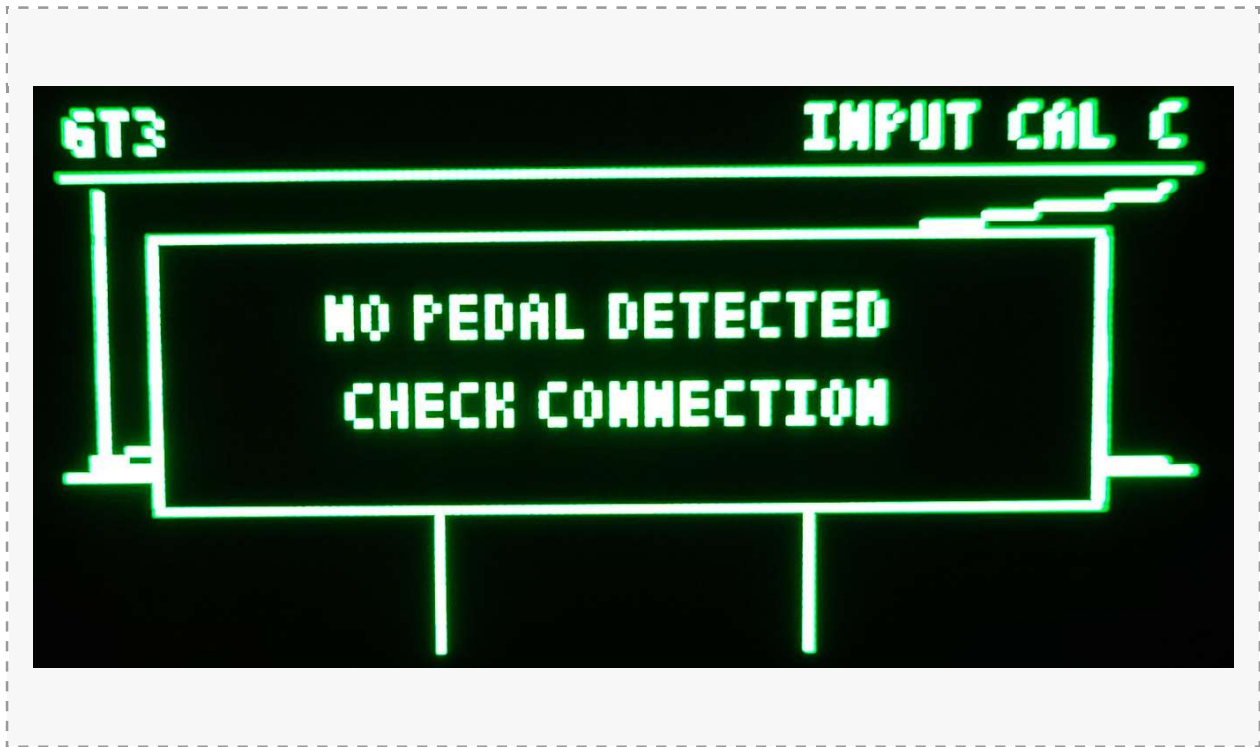


Hold 1L to confirm (the same 1.5-second progress-bar hold used elsewhere), or **click 3C** to cancel.

10. Pedal Disconnect Detection

BrakeBox continuously checks for a valid load-cell signal. If a wire comes loose or the pedal is unplugged, an overlay appears to warn that pedal is not detected. Any action on any encoder dismisses the warning, whether the pedal is connected or not.

Until a pedal is connected, a flashing INPUT indicator will appear in the top right corner. A CAL indicator will also appear, advising you to recalibrate the pedal when reconnected.



11. Troubleshooting

Issue	Action
Encoders respond intermittently or erratically	Ensure there is no input on the brake pedal while interacting with the encoders.
The input dot moves slightly with no pedal input	Re-run calibration (hold 1L). If this persists, add slight pressure to the pedal during Step 1 of calibration rather than leaving it completely untouched.
Light presses feel jittery, or the pedal ignores very light input	Re-run calibration (hold 1L); BrakeBox measures your load cell's actual noise floor during the zero step and sets the deadband accordingly.
"NO PEDAL DETECTED" appears while the pedal is connected	Check both load-cell excitation wires and the signal wire for a loose connection at the HX711.

This manual reflects the BrakeBox firmware and UI as of this revision. If your unit's screens look different from the layouts described here, your firmware may be a different version.